

Limitations

- **Requires known focal length.** CSTM needs f from EXIF; fails when metadata is stripped (social media), cropped, or unavailable. Newer methods (Depth Pro, UniDepth) predict f internally.
- **No temporal consistency.** Processes each frame independently; produces depth flickering in video. Video Depth Anything solves this with a decoder head swap.
- **Supervised ViT backbone.** Already behind DINOv2 self-supervised features when v2 launched. Tight CSTM coupling makes backbone swaps non-trivial.
- **Boundary quality.** F1 of 0.024 on natural images vs Depth Pro's 0.173. Blurry edges on thin structures.
- **Crop size sensitivity.** Published benchmarks use per-domain crop selection, violating zero-shot premise. Auto crop degrades performance.
- **Speed.** ViT-Giant2 with 8 GRU iterations too slow for real-time. Depth Pro: 0.3s with 50% fewer params.
- **Normal data gap.** Only 20K outdoor normal labels vs 16M depth labels. Normals bootstrapped from noisy depth gradients.